Pahirathan Nithilan

AI/ML Engineer

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Nithilan | Colombo

PROFILE

Enthusiastic professional with experience in developing and deploying machine learning models. Proficient in TensorFlow, PyTorch, and data analysis, with hands-on experience in data visualization and SQL. Skilled in computer vision, NLP, and creating user-friendly solutions. Committed to continuous learning and growth.

EDUCATION

Bachelor of Business Science
University of Moratuwa
2021 – present

Full-Stack Web Development Diploma (FSWD)
Uki Coding School
2019 – 2020

PROFESSIONAL EXPERIENCE

Junior AI/ML Engineer
M Data Zone SA Pvt (Ltd)
01/2025 – present

Machine Learning intern
EPR GROUPERS
08/2024 – 01/2025
I worked on advanced machine learning solutions for image recognition, voice recognition, and natural language processing (NLP) using PyTorch and TensorFlow. I conducted data analytics and statistical analysis to derive insights, improve processes, and enhance model accuracy. Using Python and OpenCV, I focused on data preprocessing, feature extraction, and deployment of robust AI models. My role included collaborating with cross-functional teams to design scalable solutions, improving performance and reliability across projects.

Co-Founder & CEO
Workwith.lk
12/2024 – present
Currently building an AI-powered platform to offer A-to-Z services. Implementing advanced AI technologies, including computer vision and natural language processing, to deliver quick, reliable, and user-friendly solutions.

PROJECTS

Machine Learning Projects
- Big Mart Sales Prediction: Implemented a sales prediction model for Big Mart items with Python and Flask. [GitHub Link]
- Customer Segmentation: Applied K-Means Clustering to group customers based on spending and income. [GitHub Link]
- Check out my Portfolio page for more exciting ML projects

Deep Learning Projects
- Digit Prediction: TensorFlow model to identify handwritten digits with a Flask interface. [GitHub]
- Object Recognition: ResNet50 model for classifying images into 10 categories, using Flask and TensorFlow. [GitHub]
- Sentiment Analysis on IMDB Reviews: LSTM model for analyzing movie review sentiments, integrated into a Flask app. [GitHub]
- Check out my portfolio page for more Deep Learning projects.

LLM & NLP Projects
- LankaMate AI Chatbot : Advanced chatbot application utilizing Gemini 1.5 for conversational interaction, text embeddings, and image insights. Built with Python and Streamlit. [GitHub]
- DocChat Lanka: System for uploading and processing documents (PDFs) with Meta LLAMA 3.1 and Groq hardware. Allows querying of document content with a history feature. [GitHub]
- Check out my portfolio page for more LLM & NLP projects.

Generative AI Projects
- Text-to-Image Generator: Generates images from text prompts using FastAPI and Stable Diffusion. Supports custom and negative prompts. [GitHub]
- AI Chat Generator – DialoGPT: Conversational AI using Microsoft's DialoGPT-large. Built with FastAPI and Hugging Face Transformers for dynamic interactions. [GitHub]
- Paraphrase Text Using Transformer: Generates paraphrased text with Google Pegasus. Features a user-friendly interface with FastAPI and Bootstrap. [GitHub]
- Check out my portfolio page for more Generative AI projects.

Freelancer
Full-Stack Web Developer (Trainee)
2019 – 2020
Designed and developed dynamic web applications using HTML5, CSS3, JavaScript, Bootstrap, Node.js, and MongoDB. Contributed to both front-end and back-end development, ensuring responsive design and smooth functionality for user-focused websites.

SKILLS

Technical Skills
- Programming: Python, Node.js, HTML
- Machine Learning: TensorFlow, PyTorch, Scikit-learn, CNN, RNN, LSTM
- Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau
- Natural Language Processing (NLP): SpaCy, NLTK, Hugging Face Transformers,GPT
- Computer Vision: OpenCV, YOLO
- Generative AI: DialoGPT, Text-to-Image Models, Conversational AI
- Databases: MySQL, PostgreSQL, MongoDB, SQLite
- Cloud Computing: AWS ,GCP, Microsoft Azure

Soft Skills
- Teamwork and Collaboration
- Effective Communication
- Problem-solving and Critical Thinking
- Adaptability
- Time Management

LANGUAGES

Tamil - Native

English - Professional Working Proficiency

REFERENCES

Dr. Amali Wijekoon, *Senior Lecturer,*
Faculty of Business - University of Moratuwa
amaliw@uom.lk, 0704381109

Dr.(Ms) Harishani Liyanage, *Senior Lecturer,*
Faculty of Business - University of Moratuwa
harishani@uom.lk

Computer Vision Projects
- Facial Landmark Recognition with TensorFlow.JS and React: Detects key facial landmarks in real-time using TensorFlow.JS and React. Identifies features like eyes, nose, and mouth for facial analysis. [GitHub]
- Object Detection App with TensorFlow.js and React.js: Detects objects in real-time using TensorFlow.js and the COCO-SSD model, recognizing 80 different classes of objects. [GitHub]
- Check out my portfolio page for more CV projects.

Final Year Research and Development Project
OREL Corporation
- Solving issues in manual MRP Process by implementing SAP MRP in LED Manufacturing of Orange Electric Factory.

University Campus Navigation System
University of Moratuwa
- Developed a QR code-based navigation system using Situm and Mapwize to improve building and room location for students and staff.

EXTRACURRICULAR ACTIVITIES

Clubs
- Member of AIESEC University of Moratuwa
- Member of Hiking Club University of Moratuwa
- Member of BPM Club University of Moratuwa
- Member of Yarl IT Hub Jaffna

INTERESTS

Personal Interests
- Vlogs
- Artificial Intelligence and Machine Learning
- Fitness
- Science Fiction and Technology
- Traveling